Improving CPU Efficiency for Data-Intensive Applications through Cache-Aware Spill Optimization

Results: Spill Ratios and Cache Miss Rates of the SPEC CPU2017 `test` input

gem5 TimingSimpleCPU · RISC-V (RV64GC) · five cache configurations · metric schema 10

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1. Overview

This report gives the results of the SPEC CPU2017 `test` input (full simulations of every run) with metric schema 10: spills are detected from runtime effective addresses alone (runtime-detected stack store–load reuse, spill-like memory traffic; *derleyicinin spill kodu olmak zorunda değil*). Every counter and formula is defined in `counter_reference_schema10.pdf` (`general_reports/definitions/`). Data as of 2026-11-12 11:16.

The same program runs on five cache configurations of equal capacity (L1D 4 KiB, L2 32 KiB, 64 B lines); only the organisation changes. The L1I is the same everywhere: 32 KiB, 2-way (256 sets), LRU.

Table 1. Cache configurations, from the least to the most associative (*en az ilişkiliden en çok ilişkiliye*).

Configuration	L1D	L2	Runs done
`small-dm`	4 KiB, direct-mapped (64 sets)	32 KiB, direct-mapped (512 sets)	34 / 35
`small-2way`	4 KiB, 2-way (32 sets), LRU	32 KiB, 8-way (64 sets), LRU	34 / 35
`small-4way`	4 KiB, 4-way (16 sets), LRU	32 KiB, 8-way (64 sets), LRU	34 / 35
`small-8way`	4 KiB, 8-way (8 sets), LRU	32 KiB, 8-way (64 sets), LRU	34 / 35
`small-fa`	4 KiB, fully associative (64 lines), LRU (FALRU)	32 KiB, fully associative (512 lines), LRU (FALRU)	34 / 35

Not finished yet, shown as —: `x264` (running in all configurations).

How the report is built: Tables 2 and 3 describe the program and do not depend on the cache, so they are shown once. Tables 4–8 give the miss rates, one table per configuration. The last page compares two benchmarks across the configurations. All ratios are percentages: 18.45 means 18.45 of every 100 (*yüzde değerler*).

2. Program counters

The program counters do not depend on the cache (*cache'ten bağımsız*). Tables 2 and 3 show the values of `small-2way`. In 26 runs the counters are identical in all five configurations. In 8 runs (`namd`, `lbm`, `imagick`, `blender`, `povray`, `omnetpp`, `wrf`, `cam4`) they differ by at most 1.8e-05 of the instructions; no ratio of Table 3 moves by more than 0.001 percentage points, below the two decimals shown. The likely causes are the run paths, whose length depends on the configuration name and which the program sees in its arguments, and the simulated time that some programs read.

Table 2. Instruction counts, in millions (*milyon komut*).

run	total insts	total load	total store	ROI insts	ROI load	ROI store	spill store	spill event	spill load
cactuBSSN	48,961.3	12,751.7	7,055.6	48,023.3	12,477.2	6,919.1	6,559.7	6,517.7	9,516.7
parest	37,270.0	11,901.1	1,010.0	37,262.4	11,899.9	1,009.7	140.5	133.4	651.8
deepsjeng	46,318.2	7,989.1	3,486.7	46,202.7	7,989.1	3,395.2	2,462.9	2,024.3	2,878.7
leela	26,994.3	5,290.0	1,847.3	26,063.9	5,110.6	1,756.7	1,430.8	1,279.5	4,144.0

run	total insts	total load	total store	ROI insts	ROI load	ROI store	spill store	spill event	spill load
namd	33,474.3	8,697.1	2,146.7	31,914.8	8,380.5	1,945.5	345.7	336.9	665.8
lbm	6,841.3	1,348.3	840.9	6,506.8	1,276.8	775.7	274.1	274.1	354.9
nab	6,706.0	1,526.2	322.9	6,691.5	1,523.1	321.6	159.2	155.6	448.5
xz	2,947.4	449.7	218.4	2,014.4	367.7	183.5	42.6	41.5	51.5
xz_4m_l1	2,934.5	448.7	210.4	2,001.5	366.7	175.5	38.4	37.4	47.0
xz_4m_l2	3,016.2	461.7	206.1	2,083.1	379.7	171.2	37.1	36.2	45.6
xz_4m_l3e	11,506.6	1,933.5	424.3	10,573.5	1,851.5	389.4	189.3	147.9	266.6
xz_4m_l4	4,945.0	791.1	283.4	4,012.0	709.1	248.5	95.1	76.6	115.9
xz_4m_l4e	13,048.8	2,230.7	434.4	12,115.7	2,148.7	399.5	194.3	151.2	277.0
xz_1m_l0	1,515.8	200.9	100.9	711.3	129.4	69.2	16.3	15.7	19.5
xz_1m_l1	1,524.4	202.6	100.7	719.8	131.0	69.0	15.6	15.1	18.9
xz_1m_l2	1,529.2	203.3	100.5	724.7	131.7	68.8	15.4	14.8	18.7
xz_1m_l3e	2,798.9	415.6	148.3	1,994.4	344.0	116.6	48.2	39.2	61.6
xz_1m_l4	1,880.6	266.5	125.1	1,076.1	194.9	93.4	32.0	26.5	37.0
xz_1m_l4e	3,003.3	452.9	150.1	2,198.8	381.3	118.4	48.9	39.7	63.2
imagick	118.6	20.7	9.0	117.7	20.6	9.0	2.5	2.5	4.0
blender	1,192.0	267.2	125.4	994.4	233.6	116.5	81.9	66.3	103.2
povray	2,242.3	716.5	288.1	2,241.0	716.3	288.0	200.3	183.1	260.9
omnetpp	12,429.4	2,794.3	1,491.5	12,426.3	2,793.8	1,491.4	968.3	909.1	1,018.3
xalancbmk	401.2	106.8	39.2	395.9	106.0	39.0	20.1	19.4	21.9
x264	—	—	—	—	—	—	—	—	—
gcc	20.8	3.7	2.1	16.0	3.0	2.0	1.2	1.1	1.2
mcf	24,622.6	8,151.8	1,327.9	24,427.0	8,113.0	1,300.6	705.2	578.7	2,193.1
perlbench_makerand	2,402.4	770.9	380.8	2,400.7	770.6	380.6	224.6	222.7	222.7
bwaves_1	17,317.2	4,937.6	798.3	17,288.4	4,937.5	771.8	145.7	145.5	467.5
bwaves_2	17,316.3	4,937.4	798.3	17,287.4	4,937.3	771.8	145.7	145.5	467.5
exchange2	57,667.5	14,464.6	10,703.4	57,667.1	14,464.5	10,703.4	2,460.7	2,087.5	3,098.3
fotonik3d	40,946.3	8,039.2	5,838.2	3,098.4	840.0	303.4	28.8	28.5	92.2
roms	30,434.9	8,337.9	2,040.4	28,955.8	7,869.3	1,964.4	212.0	203.5	1,218.3
wrf	85,116.3	18,764.3	5,413.3	84,396.9	18,603.8	5,344.4	624.1	544.6	1,168.4
cam4	141,281.7	29,915.0	9,290.9	33,466.8	6,418.1	2,815.2	1,513.3	1,021.0	1,939.6
total	690,725.8	159,788.8	57,759.3	528,070.3	126,620.5	44,528.2	19,480.6	17,522.5	31,960.2

total insts, load, store	The whole simulation: every committed instruction, load, store (*tüm simülasyon*).
ROI insts, load, store	Only between `m5_work_begin` and `m5_work_end` (*sadece ROI içinde*). LR counts as a load and SC as a store; an AMO is neither.
spill store	`roi_spill_store`: stores inside the ROI to the process stack, not atomic, no fault; read back or not (*spill koşullarını sağlayan store'lar, geri okunsun okunmasın*).
spill event	`roi_spill_event`: spill stores read back at least once; one per store, at its first reload (*en az bir kez geri okunan spill store'lar*).
spill load	`roi_spill_load`: loads that read a spill store back; repeats count, so it can exceed the events (*spill store'u geri okuyan load'lar, tekrarlar dahil*).
total	Sum over the completed runs (*tamamlanan koşuların toplamı*).

3. Program ratios

These, too, do not depend on the cache (*cache'ten bağımsız*).

Table 3. Program ratios, in percent (×100).

run	roi/ total	load/ roi	store/ roi	spill/ roi	spill ld/ load	spill st/ store	spill ev/ store	spill ev/ spill st	spill/ mem
cactuBSSN	98.08	25.98	14.41	33.48	76.27	94.81	94.20	99.36	82.88
parest	99.98	31.94	2.71	2.13	5.48	13.92	13.21	94.96	6.14
deepsjeng	99.75	17.29	7.35	11.56	36.03	72.54	59.62	82.19	46.92
leela	96.55	19.61	6.74	21.39	81.09	81.45	72.83	89.42	81.18
namd	95.34	26.26	6.10	3.17	7.95	17.77	17.32	97.47	9.80
lbm	95.11	19.62	11.92	9.67	27.79	35.34	35.34	100.00	30.65
nab	99.78	22.76	4.81	9.08	29.45	49.52	48.39	97.71	32.95
xz	68.34	18.26	9.11	4.67	14.00	23.20	22.62	97.49	17.06

run	roi/ total	load/ roi	store/ roi	spill/ roi	spill ld/ load	spill st/ store	spill ev/ store	spill ev/ spill st	spill/ mem
xz_4m_l1	68.20	18.32	8.77	4.27	12.82	21.88	21.31	97.38	15.75
xz_4m_l2	69.06	18.23	8.22	3.97	12.02	21.69	21.12	97.38	15.02
xz_4m_l3e	91.89	17.51	3.68	4.31	14.40	48.62	37.97	78.10	20.35
xz_4m_l4	81.13	17.68	6.19	5.26	16.35	38.29	30.84	80.54	22.04
xz_4m_l4e	92.85	17.74	3.30	3.89	12.89	48.64	37.86	77.83	18.49
xz_1m_l0	46.92	18.19	9.73	5.04	15.11	23.53	22.73	96.60	18.04
xz_1m_l1	47.22	18.20	9.59	4.80	14.43	22.66	21.87	96.54	17.27
xz_1m_l2	47.39	18.18	9.49	4.70	14.16	22.36	21.58	96.49	16.97
xz_1m_l3e	71.26	17.25	5.85	5.51	17.92	41.28	33.62	81.43	23.84
xz_1m_l4	57.22	18.11	8.68	6.42	19.00	34.25	28.34	82.74	23.94
xz_1m_l4e	73.21	17.34	5.38	5.10	16.58	41.28	33.53	81.23	22.43
imagick	99.27	17.51	7.63	5.48	19.31	27.54	27.30	99.14	21.81
blender	83.43	23.49	11.71	18.61	44.16	70.31	56.87	80.88	52.87
povray	99.94	31.96	12.85	20.58	36.43	69.54	63.57	91.41	45.92
omnetpp	99.97	22.48	12.00	15.99	36.45	64.93	60.96	93.88	46.36
xalancbmk	98.67	26.78	9.85	10.61	20.68	51.51	49.70	96.47	28.97
x264	—	—	—	—	—	—	—	—	—
gcc	77.03	18.45	12.20	15.19	41.69	61.46	57.73	93.93	49.56
mcf	99.21	33.21	5.32	11.87	27.03	54.22	44.50	82.07	30.79
perlbench_makerand	99.93	32.10	15.85	18.63	28.90	59.01	58.51	99.15	38.86
bwaves_1	99.83	28.56	4.46	3.55	9.47	18.87	18.85	99.87	10.74
bwaves_2	99.83	28.56	4.46	3.55	9.47	18.87	18.85	99.87	10.74
exchange2	100.00	25.08	18.56	9.64	21.42	22.99	19.50	84.83	22.09
fotonik3d	7.57	27.11	9.79	3.91	10.98	9.51	9.40	98.92	10.59
roms	95.14	27.18	6.78	4.94	15.48	10.79	10.36	96.02	14.54
wrf	99.15	22.04	6.33	2.12	6.28	11.68	10.19	87.26	7.49
cam4	23.69	19.18	8.41	10.32	30.22	53.75	36.27	67.47	37.40
pooled	76.45	23.98	8.43	9.74	25.24	43.75	39.35	89.95	30.06
mean	81.82	22.42	8.48	8.92	23.58	39.94	35.79	91.06	27.95

roi/total	`roi_total_insts` / `total_insts`: share of the program that was measured (*programın ne kadarı ölçüldü*).
load/roi	`roi_load` / `roi_total_insts`: loads among the ROI instructions (*ROI komutlarının yüzde kaçı load*).
store/roi	`roi_store` / `roi_total_insts`: stores among the ROI instructions (*yüzde kaçı store*).
spill/roi	(`roi_spill_load` + `roi_spill_store`) / `roi_total_insts`: **the main metric**, spill-like instructions (*ana metrik: stack'e yazma veya stack'ten geri okuma payı*).
spill ld/load	`roi_spill_load` / `roi_load`: loads that read a stack-written value back (*load'ların yüzde kaçı stack'ten geri okuma*).
spill st/store	`roi_spill_store` / `roi_store`: stores to the stack, read back or not (*store'ların yüzde kaçı stack'e*).
spill ev/store	`roi_spill_event` / `roi_store`: stores whose value is read later (*store'ların yüzde kaçı sonra okunan değer*).
spill ev/spill st	`roi_spill_event` / `roi_spill_store`: reuse rate of the stack writes (*stack'e yazılanların yüzde kaçı geri kullanıldı*).
spill/mem	(`roi_spill_load` + `roi_spill_store`) / (`roi_load` + `roi_store`): spill-like share of the memory instructions (*bellek trafiğinin yüzde kaçı spill-like*).
pooled	$\sum$ numerators / $\sum$ denominators over the completed runs: large runs dominate (*bütün koşular birlikte; büyük koşular baskın*).
mean	Average of the per-run percentages: every run weighs the same (*koşu değerlerinin ortalaması; her koşu eşit ağırlıkta*).

4. Cache miss rates by configuration

These depend on the configuration: the program and its instructions are the same, only the hits and misses change (*konfigürasyona göre değişir*). An *access* is one request of a load or store to a cache; an instruction that crosses a cache line makes two (*cache'e yapılan tek istek*). A *hit* ends at that level, a *miss* goes one level down (L1D → L2 → DRAM). The caches are write-allocate: a store miss first fetches the line. An L2 rate is *local*: its denominator is the L1D misses of that kind, not all accesses (*L2 oranının paydası L1D miss'leridir*). Each of Tables 4–8 is one configuration; the columns are the same in all of them.

`small-dm`

L1D: 4 KiB, direct-mapped (64 sets); L2: 32 KiB, direct-mapped (512 sets); 34 of 35 runs completed.

Table 4. Miss rates of `small-dm`, in percent of the accesses (*access'lerin yüzdesi*).

	L1D miss rate (%)							L2 local miss rate (%)				
run	all	spill ld	other ld	spill st	spill ev.	unread st	other st	all	spill ld	spill st	spill ev.	unread st
cactuBSSN	17.36	8.49	62.01	7.51	7.55	0.98	64.02	48.80	16.36	11.81	11.81	10.36
parest	15.36	9.23	16.72	5.78	5.82	4.99	3.86	47.58	11.07	7.21	6.75	17.30
deepsjeng	13.69	5.12	22.44	7.12	6.41	10.36	9.51	19.74	8.37	18.25	19.14	15.70
leela	8.96	7.13	20.94	6.84	6.20	12.23	5.89	17.75	2.77	7.37	6.24	12.25
namd	15.58	12.08	18.89	2.49	2.46	3.68	3.90	37.33	9.25	20.16	20.12	21.18
lbm	27.22	1.63	13.94	5.46	5.46	0.00	81.64	75.06	5.17	6.13	6.13	—
nab	10.79	1.73	17.24	1.79	1.66	7.39	1.93	15.30	3.84	4.09	3.52	9.54
xz	8.70	2.63	11.50	3.53	3.62	0.03	6.21	28.28	11.42	6.88	6.87	28.62
xz_4m_l1	10.08	2.96	12.69	3.84	3.72	8.23	8.19	36.30	11.62	7.92	8.29	1.57
xz_4m_l2	11.65	3.40	14.86	4.55	4.46	8.05	8.42	42.38	11.27	8.65	8.99	1.62
xz_4m_l3e	9.40	3.29	11.18	3.55	3.39	4.09	9.01	25.69	3.92	3.79	4.10	2.86
xz_4m_l4	10.81	3.19	13.89	3.38	3.23	4.01	9.28	33.53	6.66	6.50	7.36	3.63
xz_4m_l4e	9.17	3.40	10.60	3.64	3.50	4.09	9.11	24.56	3.88	3.87	4.19	2.90
xz_1m_l0	7.98	2.35	11.48	3.30	3.08	9.82	4.23	29.63	10.87	6.26	6.81	1.33
xz_1m_l1	8.79	2.39	12.22	3.62	3.35	11.03	5.39	33.78	11.04	6.29	6.86	1.53
xz_1m_l2	8.92	2.42	12.36	3.75	3.52	10.09	5.40	34.53	11.27	6.62	7.16	1.45
xz_1m_l3e	8.68	2.54	11.58	3.72	3.49	4.69	5.76	23.89	4.53	3.24	3.66	1.89
xz_1m_l4	8.85	2.43	12.59	4.04	3.76	5.37	5.59	28.51	6.97	4.12	4.79	1.88
xz_1m_l4e	8.88	2.62	11.58	3.77	3.56	4.70	5.83	22.40	4.38	3.22	3.63	1.87
imagick	7.43	8.54	7.46	0.67	0.54	16.70	9.25	47.92	4.75	52.56	46.48	74.97
blender	9.92	4.78	16.01	6.25	5.47	9.55	10.94	31.78	8.99	11.68	10.84	13.73
povray	13.66	4.34	22.83	5.54	5.06	10.64	12.34	37.82	14.60	23.38	18.77	46.74
omnetpp	15.05	3.08	29.43	4.16	3.75	10.55	9.72	36.64	16.65	10.40	8.31	21.78
xalancbmk	15.16	6.96	21.33	6.76	6.76	6.81	6.16	41.74	17.74	17.32	17.74	5.91
x264	—	—	—	—	—	—	—	—	—	—	—	—
gcc	10.14	3.83	19.44	3.21	3.07	5.34	10.21	32.44	2.73	6.08	5.99	6.91
mcf	28.22	8.02	39.58	4.46	4.44	4.54	17.83	72.37	22.46	21.23	23.79	9.76
perlbench_makerand	15.27	14.53	21.29	4.25	4.27	1.44	11.04	0.34	0.08	0.31	0.31	0.00
bwaves_1	44.12	5.35	49.10	1.37	1.37	0.77	47.46	27.18	19.23	2.42	2.39	44.33
bwaves_2	44.12	5.35	49.10	1.37	1.37	0.77	47.46	27.18	19.22	2.42	2.39	44.35
exchange2	4.01	3.38	5.33	3.77	3.88	3.12	2.49	1.04	0.20	0.64	0.60	0.97
fotonik3d	22.43	10.11	29.58	0.62	0.58	4.24	9.39	38.96	11.90	13.35	11.11	41.20
roms	30.01	7.03	37.33	3.73	3.70	4.33	21.39	38.90	15.90	7.89	7.74	10.93
wrf	26.37	10.75	28.92	5.68	5.05	10.02	23.55	26.50	12.85	13.24	12.38	16.19
cam4	14.02	8.49	16.42	9.71	7.59	14.11	18.99	44.30	23.26	44.64	38.92	51.02
pooled	18.29	6.99	25.52	6.23	5.93	8.98	14.76	36.87	13.22	15.48	13.33	28.22
mean	15.32	5.40	20.94	4.21	3.97	6.38	15.04	33.24	10.15	10.88	10.42	15.95

L1D: all	L1D demand misses / L1D demand accesses: every load and store (*bütün load ve store'lar*).
spill ld	spill-load misses / spill-load accesses (*spill load'lar*).
other ld	(`ReadReq` misses − spill-load misses) / (`ReadReq` accesses − spill-load accesses): the loads that are not spill loads (*spill olmayan load'lar*); read it next to spill ld.
spill st	spill-store misses / spill-store accesses: every spill store (*bütün spill store'lar*).
spill ev.	spill-event misses / spill-event accesses: the spill stores read back (*geri okunan spill store'lar*).
unread st	(spill-store − spill-event misses) / (spill-store − spill-event accesses): the spill stores never read back (*hiç okunmayan spill store'lar*).
other st	(`WriteReq` misses − spill-store misses) / (`WriteReq` accesses − spill-store accesses): the stores that are not spill stores (*spill olmayan store'lar*).
L2 local: all	L2 demand misses / L2 demand accesses: the L1D and L1I misses together (*L2'ye gelen bütün istekler*).
L2 local: the rest	The same kinds at L2; the denominator is that kind's L1D misses, so it can be small: read it with care (*payda o türün L1D miss'leri; küçük olabilir*).
pooled	$\sum$ numerators / $\sum$ denominators over the completed runs: large runs dominate (*bütün koşular birlikte; büyük koşular baskın*).
mean	Average of the per-run percentages: every run weighs the same (*koşu değerlerinin ortalaması; her koşu eşit ağırlıkta*).
—	Run not finished, or no access of that kind (*bitmemiş koşu ya da o türde access yok*).

`small-2way`

L1D: 4 KiB, 2-way (32 sets), LRU; L2: 32 KiB, 8-way (64 sets), LRU; 34 of 35 runs completed.

Table 5. Miss rates of `small-2way`, in percent of the accesses (*access'lerin yüzdesi*).

	L1D miss rate (%)							L2 local miss rate (%)				
run	all	spill ld	other ld	spill st	spill ev.	unread st	other st	all	spill ld	spill st	spill ev.	unread st
cactuBSSN	15.76	7.84	57.84	6.33	6.34	3.92	51.15	64.32	16.58	20.57	20.64	2.38
parest	11.59	3.74	12.85	2.08	2.03	2.98	2.84	57.67	2.53	7.18	6.18	19.99
deepsjeng	9.98	2.96	17.03	5.26	4.77	7.48	5.48	6.89	2.52	14.76	16.12	10.75
leela	6.90	5.71	16.59	4.15	3.32	11.21	5.37	20.72	1.62	12.21	12.12	12.43
namd	12.89	5.48	16.11	1.32	1.30	2.03	2.90	25.71	4.39	18.30	18.01	25.56
lbm	23.74	0.42	8.74	2.26	2.26	0.00	79.52	88.75	0.01	3.10	3.10	—
nab	6.91	1.33	10.90	1.91	1.88	3.04	0.89	4.63	0.32	1.07	1.07	1.12
xz	6.50	1.99	9.07	2.51	2.58	0.03	3.57	25.90	11.40	8.28	8.28	17.39
xz_4m_l1	8.00	2.21	10.27	3.23	3.21	4.01	6.02	35.15	11.03	7.27	7.50	0.34
xz_4m_l2	9.80	2.48	12.82	3.64	3.64	3.89	6.45	42.07	9.76	6.85	7.04	0.35
xz_4m_l3e	6.92	1.96	8.38	2.83	2.84	2.81	5.79	19.19	2.15	1.78	2.10	0.66
xz_4m_l4	8.37	2.10	10.90	3.21	2.93	4.38	6.51	30.25	5.14	3.74	4.68	1.12
xz_4m_l4e	6.70	2.06	7.88	2.91	2.95	2.78	5.84	17.71	1.96	1.66	1.94	0.62
xz_1m_l0	5.94	1.14	9.11	2.30	2.23	4.46	2.24	29.24	15.62	7.40	7.89	0.34
xz_1m_l1	6.73	1.25	9.84	2.65	2.55	5.26	3.32	33.78	13.86	6.83	7.32	0.20
xz_1m_l2	6.93	1.26	10.11	2.71	2.63	4.70	3.38	34.23	14.36	6.74	7.16	0.24
xz_1m_l3e	6.21	1.39	8.40	3.26	2.64	5.98	3.62	18.66	3.60	1.89	2.81	0.11
xz_1m_l4	6.83	1.35	9.94	3.88	2.84	8.85	3.70	24.25	7.31	2.51	4.03	0.17
xz_1m_l4e	6.17	1.45	8.11	3.30	2.71	5.87	3.65	17.52	3.39	1.82	2.67	0.11
imagick	4.42	0.88	4.53	0.57	0.44	15.79	7.77	81.36	21.05	33.95	25.22	61.63
blender	5.34	1.78	8.49	3.99	3.38	6.56	7.28	19.43	0.77	1.68	0.63	3.95
povray	10.14	2.54	17.68	3.32	2.76	9.31	9.24	41.21	24.21	40.15	34.01	59.52
omnetpp	12.56	1.71	25.66	2.99	2.52	10.14	6.98	26.59	1.80	4.98	4.49	6.86
xalancbmk	13.66	5.51	19.83	5.18	5.24	3.76	4.65	33.74	5.75	13.33	13.59	3.52
x264	—	—	—	—	—	—	—	—	—	—	—	—
gcc	7.17	1.31	14.79	1.32	1.23	2.67	8.63	32.52	2.56	9.39	9.12	11.25
mcf	25.50	4.28	36.86	3.71	3.84	3.13	16.53	75.88	30.53	17.58	19.80	5.14
perlbench_makerand	5.72	0.88	10.94	0.90	0.88	2.72	1.23	0.00	0.00	0.00	0.00	0.00
bwaves_1	41.07	2.70	45.89	0.33	0.33	0.43	44.79	31.22	10.11	0.77	0.64	80.02
bwaves_2	41.07	2.70	45.89	0.33	0.33	0.43	44.79	31.22	10.11	0.77	0.64	80.10
exchange2	2.50	3.23	3.11	3.85	4.31	1.28	1.00	1.08	0.14	0.52	0.44	2.19
fotonik3d	17.86	5.94	24.32	0.25	0.20	5.12	6.13	41.35	1.19	19.82	11.14	50.92
roms	21.00	4.38	25.97	1.66	1.70	0.71	16.03	38.82	7.97	10.43	9.50	63.55
wrf	18.06	6.63	19.80	3.58	3.23	5.96	16.36	22.28	6.88	9.77	10.38	7.50
cam4	10.24	5.57	11.79	7.75	5.63	12.15	14.79	52.99	21.61	52.25	48.91	55.46
pooled	14.47	5.16	20.42	4.88	4.63	7.09	11.36	40.55	12.09	19.66	17.97	29.58
mean	12.03	2.89	16.78	2.93	2.70	4.82	12.01	33.13	8.01	10.27	9.68	17.74

L1D: all	L1D demand misses / L1D demand accesses: every load and store (*bütün load ve store'lar*).
spill ld	spill-load misses / spill-load accesses (*spill load'lar*).
other ld	(`ReadReq` misses − spill-load misses) / (`ReadReq` accesses − spill-load accesses): the loads that are not spill loads (*spill olmayan load'lar*); read it next to spill ld.
spill st	spill-store misses / spill-store accesses: every spill store (*bütün spill store'lar*).
spill ev.	spill-event misses / spill-event accesses: the spill stores read back (*geri okunan spill store'lar*).
unread st	(spill-store − spill-event misses) / (spill-store − spill-event accesses): the spill stores never read back (*hiç okunmayan spill store'lar*).
other st	(`WriteReq` misses − spill-store misses) / (`WriteReq` accesses − spill-store accesses): the stores that are not spill stores (*spill olmayan store'lar*).
L2 local: all	L2 demand misses / L2 demand accesses: the L1D and L1I misses together (*L2'ye gelen bütün istekler*).
L2 local: the rest	The same kinds at L2; the denominator is that kind's L1D misses, so it can be small: read it with care (*payda o türün L1D miss'leri; küçük olabilir*).
pooled	$\sum$ numerators / $\sum$ denominators over the completed runs: large runs dominate (*bütün koşular birlikte; büyük koşular baskın*).
mean	Average of the per-run percentages: every run weighs the same (*koşu değerlerinin ortalaması; her koşu eşit ağırlıkta*).
—	Run not finished, or no access of that kind (*bitmemiş koşu ya da o türde access yok*).

`small-4way`

L1D: 4 KiB, 4-way (16 sets), LRU; L2: 32 KiB, 8-way (64 sets), LRU; 34 of 35 runs completed.

Table 6. Miss rates of `small-4way`, in percent of the accesses (*access'lerin yüzdesi*).

	L1D miss rate (%)							L2 local miss rate (%)				
run	all	spill ld	other ld	spill st	spill ev.	unread st	other st	all	spill ld	spill st	spill ev.	unread st
cactuBSSN	15.30	8.14	55.03	6.37	6.38	4.30	40.30	65.69	16.17	19.53	19.60	2.17
parest	10.21	1.29	11.46	1.08	1.03	2.07	2.23	65.62	7.72	13.82	12.23	28.65
deepsjeng	8.90	2.26	15.30	4.82	4.43	6.61	5.14	7.54	3.25	15.98	17.25	12.05
leela	5.85	4.76	14.55	3.25	2.62	8.58	5.34	24.20	1.95	15.59	15.32	16.29
namd	11.71	2.78	14.82	1.14	1.14	1.20	2.66	28.28	8.69	21.24	20.64	43.17
lbm	21.66	0.00	7.78	0.68	0.68	0.00	73.96	97.36	100.00	10.39	10.39	—
nab	5.33	0.50	8.76	0.38	0.35	1.54	0.80	6.02	0.84	5.42	5.75	2.21
xz	5.69	1.46	8.26	1.52	1.56	0.02	2.75	29.53	15.53	13.67	13.66	25.32
xz_4m_l1	7.17	1.80	9.46	2.07	2.05	2.77	5.11	39.21	13.49	11.37	11.76	0.50
xz_4m_l2	9.13	2.22	12.22	2.57	2.57	2.78	5.60	45.18	10.91	9.71	9.98	0.48
xz_4m_l3e	6.35	1.12	7.95	1.92	2.01	1.63	4.82	20.93	3.77	2.63	2.97	1.14
xz_4m_l4	7.39	1.21	9.98	1.82	1.66	2.49	5.52	34.26	8.89	6.59	8.27	1.97
xz_4m_l4e	6.20	1.22	7.51	2.03	2.14	1.62	4.87	19.17	3.31	2.39	2.67	1.07
xz_1m_l0	5.14	0.85	8.23	1.15	1.09	2.77	1.55	33.71	20.99	14.87	16.15	0.55
xz_1m_l1	5.81	0.96	8.87	1.33	1.25	3.65	2.44	39.07	18.15	13.57	14.97	0.29
xz_1m_l2	6.07	1.00	9.21	1.43	1.36	3.25	2.52	39.07	18.09	12.79	13.87	0.34
xz_1m_l3e	5.46	0.76	7.78	1.86	1.41	3.84	2.65	21.23	6.58	3.30	5.24	0.18
xz_1m_l4	5.83	0.78	8.99	2.00	1.21	5.80	2.73	28.43	12.68	4.86	9.48	0.26
xz_1m_l4e	5.47	0.82	7.55	1.92	1.49	3.78	2.68	19.76	5.98	3.12	4.85	0.16
imagick	4.20	0.25	4.33	0.53	0.40	15.08	7.67	85.57	72.96	36.50	27.37	64.38
blender	5.06	1.63	7.94	3.96	3.38	6.42	7.05	20.38	0.83	1.69	0.65	4.01
povray	9.13	2.08	16.18	2.68	2.09	9.00	8.25	45.58	29.53	50.12	45.30	62.04
omnetpp	11.85	1.25	24.56	2.57	2.10	9.89	6.51	28.01	2.46	5.93	5.51	7.32
xalancbmk	13.28	4.52	19.52	4.82	4.93	1.71	4.65	34.60	7.25	14.27	14.35	7.80
x264	—	—	—	—	—	—	—	—	—	—	—	—
gcc	6.10	0.58	12.74	0.92	0.82	2.49	8.22	36.94	5.70	13.47	13.77	11.94
mcf	24.85	3.52	36.16	3.30	3.51	2.37	16.42	77.84	37.14	19.71	21.63	6.77
perlbench_makerand	3.76	0.00	7.17	0.06	0.00	6.66	2.45	0.00	0.11	0.01	0.12	0.00
bwaves_1	40.95	3.81	45.81	0.02	0.02	0.38	43.46	31.23	6.87	11.34	9.59	88.97
bwaves_2	40.95	3.81	45.81	0.02	0.02	0.38	43.46	31.23	6.87	11.34	9.59	89.32
exchange2	2.01	1.91	2.63	2.74	2.93	1.69	0.97	1.35	0.24	0.75	0.65	1.66
fotonik3d	11.96	4.10	15.99	0.20	0.15	4.79	4.86	61.79	1.72	25.72	15.38	54.49
roms	14.94	3.01	18.14	0.89	0.90	0.68	12.75	53.16	11.67	19.33	17.83	66.81
wrf	11.55	4.95	12.40	2.81	2.58	4.41	11.20	34.19	9.26	12.42	12.99	10.12
cam4	9.07	4.56	10.31	7.05	5.06	11.17	13.90	59.49	26.39	57.37	54.39	60.17
pooled	12.54	4.56	17.67	4.40	4.19	6.31	9.65	46.20	13.77	21.35	19.36	33.20
mean	10.72	2.17	15.10	2.12	1.92	4.00	10.75	37.22	14.59	14.14	13.65	20.38

L1D: all	L1D demand misses / L1D demand accesses: every load and store (*bütün load ve store'lar*).
spill ld	spill-load misses / spill-load accesses (*spill load'lar*).
other ld	(`ReadReq` misses − spill-load misses) / (`ReadReq` accesses − spill-load accesses): the loads that are not spill loads (*spill olmayan load'lar*); read it next to spill ld.
spill st	spill-store misses / spill-store accesses: every spill store (*bütün spill store'lar*).
spill ev.	spill-event misses / spill-event accesses: the spill stores read back (*geri okunan spill store'lar*).
unread st	(spill-store − spill-event misses) / (spill-store − spill-event accesses): the spill stores never read back (*hiç okunmayan spill store'lar*).
other st	(`WriteReq` misses − spill-store misses) / (`WriteReq` accesses − spill-store accesses): the stores that are not spill stores (*spill olmayan store'lar*).
L2 local: all	L2 demand misses / L2 demand accesses: the L1D and L1I misses together (*L2'ye gelen bütün istekler*).
L2 local: the rest	The same kinds at L2; the denominator is that kind's L1D misses, so it can be small: read it with care (*payda o türün L1D miss'leri; küçük olabilir*).
pooled	$\sum$ numerators / $\sum$ denominators over the completed runs: large runs dominate (*bütün koşular birlikte; büyük koşular baskın*).
mean	Average of the per-run percentages: every run weighs the same (*koşu değerlerinin ortalaması; her koşu eşit ağırlıkta*).
—	Run not finished, or no access of that kind (*bitmemiş koşu ya da o türde access yok*).

`small-8way`

L1D: 4 KiB, 8-way (8 sets), LRU; L2: 32 KiB, 8-way (64 sets), LRU; 34 of 35 runs completed.

Table 7. Miss rates of `small-8way`, in percent of the accesses (*access'lerin yüzdesi*).

	L1D miss rate (%)							L2 local miss rate (%)				
run	all	spill ld	other ld	spill st	spill ev.	unread st	other st	all	spill ld	spill st	spill ev.	unread st
cactuBSSN	15.62	8.68	54.86	6.81	6.82	4.55	37.28	64.56	15.26	17.75	17.82	1.87
parest	9.90	0.77	11.15	0.84	0.78	1.95	2.02	67.70	13.19	17.75	16.08	30.43
deepsjeng	8.63	2.07	14.83	4.78	4.39	6.53	5.07	7.70	3.53	16.06	17.34	12.09
leela	5.16	4.03	13.48	2.77	2.22	7.42	5.37	27.19	2.30	18.27	18.04	18.83
namd	11.61	2.55	14.72	1.16	1.16	1.21	2.61	28.50	9.47	20.82	20.23	42.67
lbm	21.73	0.00	7.48	0.63	0.63	0.00	74.83	97.05	100.00	11.09	11.09	—
nab	4.07	0.18	6.77	0.23	0.21	0.82	0.78	7.87	2.32	9.05	9.49	4.14
xz	5.34	0.86	7.97	0.85	0.87	0.02	2.43	31.37	26.50	24.55	24.54	28.85
xz_4m_l1	6.79	1.17	9.10	1.27	1.30	0.15	4.87	41.37	20.77	18.52	18.54	9.29
xz_4m_l2	8.81	1.77	11.90	1.87	1.91	0.34	5.43	46.80	13.67	13.35	13.39	4.01
xz_4m_l3e	6.09	0.58	7.82	1.23	1.37	0.72	4.42	21.75	7.29	4.11	4.34	2.58
xz_4m_l4	6.79	0.67	9.35	1.00	1.07	0.72	5.12	37.27	16.00	12.03	12.87	6.87
xz_4m_l4e	5.97	0.68	7.40	1.33	1.50	0.74	4.45	19.86	5.94	3.63	3.81	2.33
xz_1m_l0	4.78	0.55	7.86	0.52	0.53	0.28	1.25	36.16	32.29	32.80	33.31	5.38
xz_1m_l1	5.39	0.64	8.42	0.64	0.65	0.18	2.13	42.00	27.34	28.31	28.54	5.72
xz_1m_l2	5.68	0.76	8.78	0.77	0.79	0.19	2.26	41.63	23.81	23.64	23.79	5.85
xz_1m_l3e	4.97	0.39	7.34	0.82	0.87	0.58	2.23	23.27	12.86	7.53	8.50	1.16
xz_1m_l4	5.05	0.42	8.08	0.68	0.69	0.65	2.31	32.77	23.41	14.27	16.66	2.27
xz_1m_l4e	5.02	0.45	7.16	0.88	0.94	0.60	2.25	21.50	10.89	6.81	7.65	1.02
imagick	4.17	0.23	4.29	0.50	0.37	15.04	7.66	86.22	79.83	38.88	29.73	64.64
blender	5.04	1.49	7.90	4.09	3.45	6.78	7.06	20.42	0.90	1.64	0.63	3.81
povray	8.70	1.81	15.54	2.50	1.88	9.09	7.80	47.73	33.95	53.73	50.15	61.62
omnetpp	11.57	1.10	23.91	2.74	2.22	10.65	6.44	28.59	2.80	5.60	5.21	6.83
xalancbmk	13.12	4.50	19.29	4.65	4.77	1.20	4.62	35.01	7.52	14.77	14.80	11.07
x264	—	—	—	—	—	—	—	—	—	—	—	—
gcc	5.73	0.41	11.96	0.77	0.69	2.09	8.11	38.80	8.05	16.05	16.44	14.10
mcf	24.64	3.36	35.91	3.18	3.39	2.22	16.41	78.48	38.85	20.50	22.39	7.23
perlbench_makerand	3.80	0.00	7.28	0.85	0.00	100.00	1.23	0.00	0.11	0.00	0.12	0.00
bwaves_1	26.02	5.47	26.51	0.02	0.02	0.38	43.92	37.38	4.30	10.94	9.24	90.59
bwaves_2	26.02	5.47	26.51	0.02	0.02	0.37	43.92	37.38	4.30	10.94	9.24	91.35
exchange2	1.34	1.63	1.74	2.40	2.48	1.95	0.37	2.01	0.29	0.85	0.77	1.45
fotonik3d	9.21	2.77	11.96	0.19	0.15	4.31	4.81	80.25	2.54	26.39	15.36	60.49
roms	11.83	2.42	13.85	0.98	0.99	0.66	12.05	67.05	14.29	17.44	16.03	68.40
wrf	6.92	4.46	7.05	2.70	2.47	4.28	7.60	55.78	9.96	12.91	13.53	10.43
cam4	8.84	4.21	10.08	6.86	4.91	10.91	13.77	60.99	28.58	58.97	56.06	61.69
pooled	10.50	4.51	14.25	4.43	4.23	6.18	8.67	52.53	13.94	20.96	18.85	33.87
mean	9.25	1.96	13.18	1.81	1.66	5.81	10.38	40.36	17.74	17.35	16.64	22.40

L1D: all	L1D demand misses / L1D demand accesses: every load and store (*bütün load ve store'lar*).
spill ld	spill-load misses / spill-load accesses (*spill load'lar*).
other ld	(`ReadReq` misses − spill-load misses) / (`ReadReq` accesses − spill-load accesses): the loads that are not spill loads (*spill olmayan load'lar*); read it next to spill ld.
spill st	spill-store misses / spill-store accesses: every spill store (*bütün spill store'lar*).
spill ev.	spill-event misses / spill-event accesses: the spill stores read back (*geri okunan spill store'lar*).
unread st	(spill-store − spill-event misses) / (spill-store − spill-event accesses): the spill stores never read back (*hiç okunmayan spill store'lar*).
other st	(`WriteReq` misses − spill-store misses) / (`WriteReq` accesses − spill-store accesses): the stores that are not spill stores (*spill olmayan store'lar*).
L2 local: all	L2 demand misses / L2 demand accesses: the L1D and L1I misses together (*L2'ye gelen bütün istekler*).
L2 local: the rest	The same kinds at L2; the denominator is that kind's L1D misses, so it can be small: read it with care (*payda o türün L1D miss'leri; küçük olabilir*).
pooled	$\sum$ numerators / $\sum$ denominators over the completed runs: large runs dominate (*bütün koşular birlikte; büyük koşular baskın*).
mean	Average of the per-run percentages: every run weighs the same (*koşu değerlerinin ortalaması; her koşu eşit ağırlıkta*).
—	Run not finished, or no access of that kind (*bitmemiş koşu ya da o türde access yok*).

`small-fa`

L1D: 4 KiB, fully associative (64 lines), LRU (FALRU); L2: 32 KiB, fully associative (512 lines), LRU (FALRU); 34 of 35 runs completed.

Table 8. Miss rates of `small-fa`, in percent of the accesses (*access'lerin yüzdesi*).

	L1D miss rate (%)							**L2 local miss rate (%)**				
run	all	spill ld	other ld	spill st	spill ev.	unread st	other st	all	spill ld	spill st	spill ev.	unread st
cactuBSSN	15.22	8.98	51.27	7.34	7.36	3.51	27.78	59.94	15.69	14.43	14.47	1.02
parest	9.72	0.59	10.96	0.66	0.58	2.12	1.99	68.60	15.87	22.92	21.79	28.72
deepsjeng	8.08	1.78	13.86	4.68	4.31	6.39	4.80	6.11	2.74	13.24	14.43	9.53
leela	5.07	4.01	13.21	2.59	2.12	6.52	5.38	27.89	2.40	20.20	19.72	21.50
namd	11.61	2.56	14.73	1.17	1.17	1.21	2.61	28.07	7.85	18.32	17.71	41.21
lbm	21.23	0.00	7.51	0.55	0.55	0.00	72.77	99.28	100.00	3.25	3.25	—
nab	3.89	0.16	6.46	0.21	0.20	0.71	0.77	8.31	2.94	10.06	10.50	4.82
xz	5.09	0.71	7.78	0.63	0.65	0.02	2.00	31.78	32.02	36.52	36.53	28.37
xz_4m_l1	6.62	0.88	9.00	0.89	0.92	0.03	4.66	41.40	27.63	28.26	28.25	45.91
xz_4m_l2	8.61	1.16	11.72	1.51	1.55	0.03	5.36	47.32	20.81	17.44	17.43	41.21
xz_4m_l3e	6.04	0.45	7.81	1.03	1.17	0.53	4.19	21.48	9.55	5.45	5.69	3.48
xz_4m_l4	6.39	0.51	8.85	0.68	0.74	0.44	4.85	38.80	21.16	17.76	18.85	10.22
xz_4m_l4e	5.90	0.54	7.37	1.14	1.30	0.57	4.22	19.54	7.49	4.72	4.93	3.01
xz_1m_l0	4.62	0.51	7.68	0.45	0.46	0.29	1.05	36.56	35.88	40.96	41.74	5.57
xz_1m_l1	5.20	0.55	8.17	0.52	0.54	0.02	1.98	42.80	31.69	36.56	36.54	43.41
xz_1m_l2	5.49	0.61	8.52	0.66	0.68	0.02	2.17	42.54	29.73	29.61	29.59	45.45
xz_1m_l3e	4.76	0.31	7.10	0.64	0.74	0.20	2.00	23.80	16.11	10.43	10.86	3.49
xz_1m_l4	4.64	0.35	7.48	0.46	0.52	0.16	2.11	34.83	28.21	22.06	22.88	9.01
xz_1m_l4e	4.81	0.37	6.93	0.70	0.81	0.23	2.01	21.91	13.33	9.24	9.67	2.75
imagick	4.16	0.22	4.28	0.49	0.36	14.94	7.65	87.38	82.76	38.96	29.85	64.38
blender	5.13	1.72	7.95	4.11	3.53	6.58	7.06	19.66	0.77	1.42	0.53	3.46
povray	8.70	1.86	15.41	2.50	1.87	9.18	8.36	50.57	34.86	61.76	57.89	70.16
omnetpp	11.53	1.01	23.85	2.81	2.28	10.86	6.33	27.53	2.70	4.71	4.79	4.45
xalancbmk	13.00	4.37	19.06	4.94	5.07	1.27	4.63	34.56	6.36	10.79	10.77	12.40
x264	—	—	—	—	—	—	—	—	—	—	—	—
gcc	5.69	0.39	11.91	0.72	0.64	1.96	8.03	38.30	6.93	16.21	16.66	13.97
mcf	24.52	3.30	35.73	3.17	3.39	2.13	16.43	78.48	39.32	18.92	20.60	6.69
perlbench_makerand	0.08	0.00	0.17	0.01	0.00	0.26	0.01	0.08	0.11	0.08	0.13	0.00
bwaves_1	7.31	0.86	7.75	0.00	0.00	0.39	10.68	95.60	17.43	67.68	64.10	88.24
bwaves_2	7.31	0.86	7.75	0.00	0.00	0.39	10.68	95.60	17.43	67.70	64.08	88.58
exchange2	1.54	1.78	2.09	2.43	2.49	2.09	0.42	1.79	0.28	0.96	0.90	1.37
fotonik3d	8.89	2.18	11.54	0.20	0.15	4.89	4.81	82.63	1.81	0.96	1.13	0.47
roms	10.76	2.14	12.40	1.04	1.05	0.66	11.72	64.31	10.39	14.85	13.44	68.68
wrf	5.88	3.94	6.01	2.61	2.39	4.12	6.33	61.76	10.61	16.01	15.49	18.07
cam4	8.80	4.10	10.09	6.67	4.79	10.57	13.84	59.48	28.54	60.45	56.85	63.84
pooled	8.92	4.40	11.97	4.55	4.40	5.88	6.56	56.35	14.56	18.89	16.46	35.16
mean	7.83	1.58	11.54	1.71	1.60	2.74	7.93	44.08	20.04	21.85	21.24	25.86

L1D: all	L1D demand misses / L1D demand accesses: every load and store (*bütün load ve store'lar*).
spill ld	spill-load misses / spill-load accesses (*spill load'lar*).
other ld	(`ReadReq` misses − spill-load misses) / (`ReadReq` accesses − spill-load accesses): the loads that are not spill loads (*spill olmayan load'lar*); read it next to spill ld.
spill st	spill-store misses / spill-store accesses: every spill store (*bütün spill store'lar*).
spill ev.	spill-event misses / spill-event accesses: the spill stores read back (*geri okunan spill store'lar*).
unread st	(spill-store − spill-event misses) / (spill-store − spill-event accesses): the spill stores never read back (*hiç okunmayan spill store'lar*).
other st	(`WriteReq` misses − spill-store misses) / (`WriteReq` accesses − spill-store accesses): the stores that are not spill stores (*spill olmayan store'lar*).
L2 local: all	L2 demand misses / L2 demand accesses: the L1D and L1I misses together (*L2'ye gelen bütün istekler*).
L2 local: the rest	The same kinds at L2; the denominator is that kind's L1D misses, so it can be small: read it with care (*payda o türün L1D miss'leri; küçük olabilir*).
pooled	$\sum$ numerators / $\sum$ denominators over the completed runs: large runs dominate (*bütün koşular birlikte; büyük koşular baskın*).
mean	Average of the per-run percentages: every run weighs the same (*koşu değerlerinin ortalaması; her koşu eşit ağırlıkta*).
—	Run not finished, or no access of that kind (*bitmemiş koşu ya da o türde access yok*).

5. Comparison of two benchmarks

To make the tables concrete, this page follows two benchmarks through the five configurations (*iki benchmark'ın konfigürasyonlar boyunca karşılaştırması*). By spill share (`spill/roi`) `cactuBSSN` ranks 1st and `parest` 33rd of the 34 completed runs of the `test` input. The rows of the cache tables go from the least to the most associative; the last row is the change from direct-mapped to fully associative, in percentage points (*yüzde puan farkı*).

Table 9. Program profile (does not depend on the cache), in percent.

	`cactuBSSN`	`parest`
ROI instructions (millions)	48,023.3	37,262.4
`spill/roi`	33.48	2.13
`spill ld/load`	76.27	5.48
`spill st/store`	94.81	13.92
`spill ev/spill st`	99.36	94.96
`spill/mem`	82.88	6.14

Table 10. L1D miss rates by configuration, in percent (columns as in Tables 4–8).

	`cactuBSSN` (%)					`parest` (%)				
config	L1D all	spill ld	other ld	spill st	other st	L1D all	spill ld	other ld	spill st	other st
`dm`	17.36	8.49	62.01	7.51	64.02	15.36	9.23	16.72	5.78	3.86
`2way`	15.76	7.84	57.84	6.33	51.15	11.59	3.74	12.85	2.08	2.84
`4way`	15.30	8.14	55.03	6.37	40.30	10.21	1.29	11.46	1.08	2.23
`8way`	15.62	8.68	54.86	6.81	37.28	9.90	0.77	11.15	0.84	2.02
`fa`	15.22	8.98	51.27	7.34	27.78	9.72	0.59	10.96	0.66	1.99
dm − fa	2.13	-0.48	10.74	0.17	36.24	5.64	8.64	5.76	5.12	1.87

Table 11. L2 local miss rates by configuration, in percent.

	`cactuBSSN` (%)			`parest` (%)		
config	L2 all	spill ld	spill st	L2 all	spill ld	spill st
`dm`	48.80	16.36	11.81	47.58	11.07	7.21
`2way`	64.32	16.58	20.57	57.67	2.53	7.18
`4way`	65.69	16.17	19.53	65.62	7.72	13.82
`8way`	64.56	15.26	17.75	67.70	13.19	17.75
`fa`	59.94	15.69	14.43	68.60	15.87	22.92
dm − fa	-11.13	0.67	-2.62	-21.02	-4.80	-15.72

Reading (*okuması*):

- **Program.** `cactuBSSN` spends 33.48% of its ROI instructions on spill-like traffic, `parest` 2.13%. That is 82.88% of the memory instructions of `cactuBSSN` and 6.14% of those of `parest` (*bellek trafiğindeki spill payı*). These values are the same in every configuration.
- **Spill loads against other loads.** Over the five configurations the spill loads of `cactuBSSN` miss L1D 7.84–8.98% of the time, its other loads 51.27–62.01%; for `parest` 0.59–9.23% and 10.96–16.72% (*spill load'lar diğer load'lardan daha az miss ediyor mu?*).
- **Associativity.** From `dm` to `fa` the spill-load miss rate of `cactuBSSN` goes 8.49% → 8.98%: it hardly changes with the organisation, which points to capacity rather than conflict misses (*çakışma değil kapasite miss'leri*). That of `parest` goes 9.23% → 0.59%: it falls sharply with associativity: in the direct-mapped cache these are mostly conflict misses (*çoğu çakışma miss'i*). The overall L1D miss rate goes 17.36% → 15.22% for `cactuBSSN` and 15.36% → 9.72% for `parest`.
- **Weight of the spill traffic.** Spill loads and spill stores make 82.9% of the L1D accesses of `cactuBSSN` but 38.0–45.2% of its L1D misses; for `parest` 6.1% of the accesses and 0.4–3.4% of the misses (*spill trafiğinin access'ler ve miss'ler içindeki payı*).
- **L2.** Of the accesses that miss L1D, L2 misses 48.80–65.69% for `cactuBSSN` and 47.58–68.60% for `parest`; the spill loads that reach L2 miss there 15.26–16.58% and 2.53–15.87%.